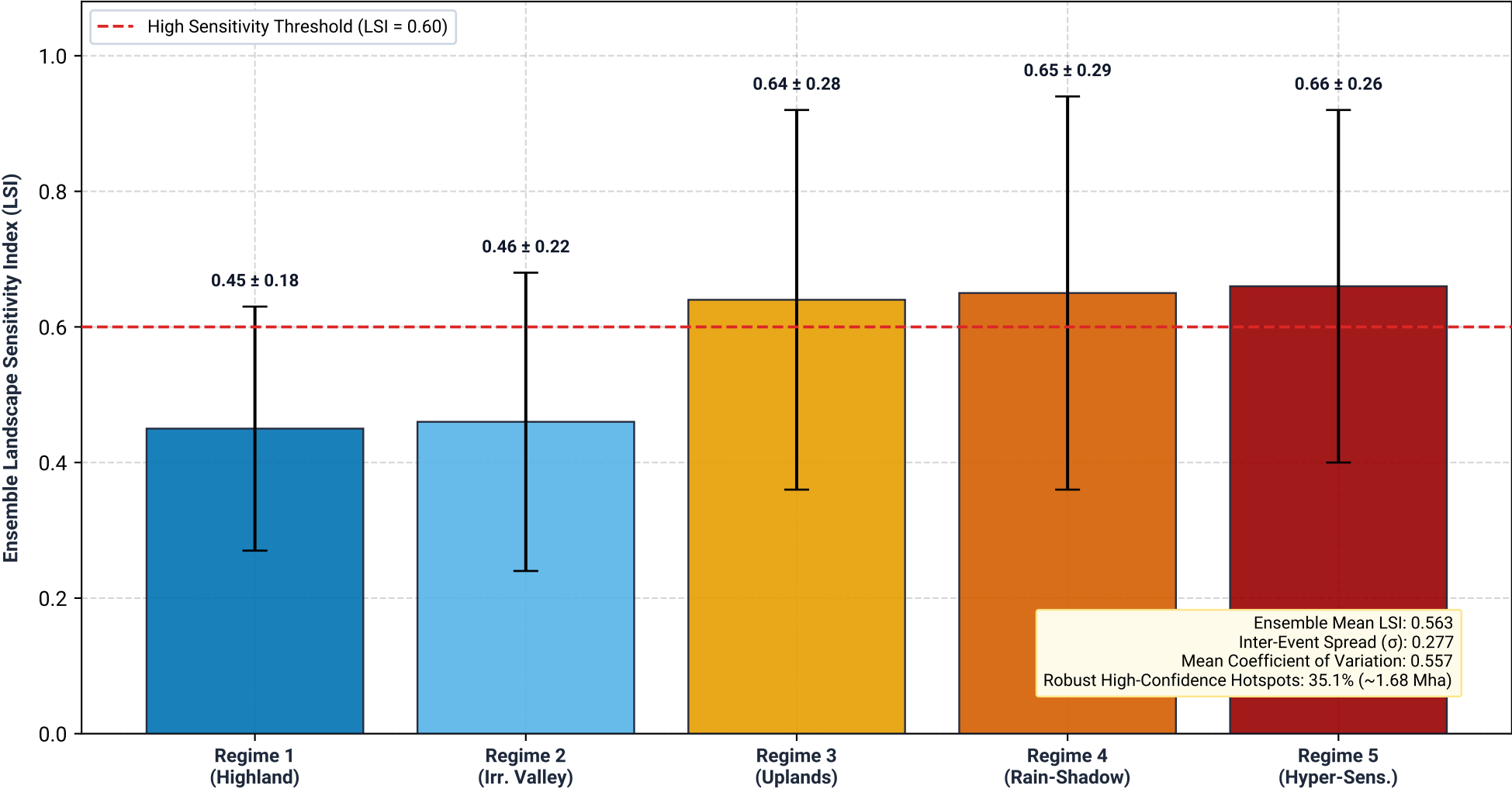


Multi-Event LSI Ensemble Uncertainty (8 Historical El Niño Episodes across 5 Biophysical Regimes)
Bar: Multi-Event Mean | Whiskers: Inter-Episode Spread ($\pm 1\sigma$) across 2002, 2004, 2006, 2009, 2014, 2015, 2018, 2023



• Multi-Episode Spread: Provincial mean LSI is 0.563 with inter-event spread $\sigma = 0.277$ (CV = 0.557) across 8 historical El Niño episodes.

• Deterministic Core: 35.1% of East Java (~1.68 Mha) exceeds LSI ≥ 0.60 across all ENSO episode types (Canonical, Modoki, and Compound IOD+).

• Stable Natural Buffers: Highlands (Regime 1) and irrigated river valleys (Regime 2) exhibit low variance (CV ≤ 0.35) and high buffering capacity.